

List of Experiments of C Programming

Basic I/O (CO1)

- 1: Write a program to print a simple message.
- 2: Write a program to display your name, roll number, course and college name.
- 3: Write a program to swap two integers using a temporary variable.
- 4: Write a program to calculate the area and circumference of a circle. Take the radius as input from the user.
- 5: Write a program to calculate the area and perimeter of a rectangle. Take the length and breadth as input from the user.
- 6: Write a program to convert temperature from Celsius to Fahrenheit and Fahrenheit to Celsius. Take the temperature as input from the user.

Control Flow (CO1, CO2)

- 7: Write a program to find the largest of three numbers.
- 8: Write a program to input a student's marks and assign a grade (A, B, C, D, F) using if-else-if ladder.
- 9: Write a program to demonstrate the use of switch-case statement.
- 10: Write a program to check whether a given number is even or odd.
- 11: Write a program to check whether a given number is positive, negative, or zero.
- 12: Write a program to check whether a given character is a vowel or consonant.

Loops (CO1, CO2)

- 13: Write a program to print all prime numbers between 1 and N, where N is entered by the user.
- 14: Write a program to find the factorial of a given number using a loop.
- 15: Write a program to generate the Fibonacci series up to N terms using a loop.
- 16: Write a program to print the multiplication table of a given number.
- 17: Write a program to reverse a given integer (e.g., 1234 becomes 4321) and check whether it is a palindrome.
- 18: Write a program to check whether a given number is an Armstrong number.
- 19: Write a program to display the following pattern:

```
*
**
***
****
*****
```

Functions (Call by Value) (CO1, CO2, CO3)

- 20:** Write a function `isPrime()` that takes an integer as an argument and returns 1 if it is prime, otherwise returns 0.
- 21:** Write a function `calculateFactorial()` that takes an integer as an argument and returns its factorial.
- 22:** Write a program to find the maximum and minimum of two numbers using user-defined functions.
- 23:** Write a program to calculate the power of a number using a user-defined function with call by value.

Functions (Call by Reference) & Recursion (CO1, CO2, CO3)

- 24:** Write a function `swap()` that swaps the values of two integers using call by reference.
- 25:** Write a recursive function to calculate the factorial of a number.
- 26:** Write a recursive function to calculate the nth Fibonacci number.

Arrays (CO1, CO2, CO3)

- 27:** Write a program to find the sum and average of elements in an array.
- 28:** Write a program to find the largest and smallest element in an integer array.
- 29:** Write a program to search for a given element in an array using Linear Search.
- 30:** Write a program to count the number of even and odd elements in an array.
- 31:** Write a program to perform addition of two matrices using two-dimensional arrays.
- 32:** Write a program to find the transpose of a matrix.
- 33:** Write a program to check whether a given matrix is symmetric or not.

Strings (CO1, CO2, CO3)

- 34:** Write a program to perform the following operations on strings without using inbuilt string functions:
1. Calculate the length of a string.
 2. Concatenate two strings.
 3. Compare two strings.
 4. Copy one string to another.
- 35:** Write a program to check whether a given string is a palindrome.
- 36:** Write a program to count the number of vowels, consonants, digits and spaces in a given string.

Pointers (CO1, CO2, CO4)

- 37:** Write a program to demonstrate the use of & and * operators to access and modify a variable.
- 38:** Write a program to access and display elements of an integer array using a pointer.
- 39:** Write a program to perform addition of two numbers using pointers.
- 40:** Write a program to demonstrate pointer arithmetic.

Structures (CO1, CO2, CO5)

- 41:** Create a Student structure with members name, roll number and marks. Write a program to read and display the data of a student.
- 42:** Create an Employee structure with members employee ID, name and salary. Write a program to read and display data for three employees.
- 43:** Write a program to demonstrate a union containing integer, float and character members.

File Handling (CO1, CO4)

- 44:** Write a program to write a simple sentence to a text file.
- 45:** Write a program to read the contents of a text file and display them on the screen.
- 46:** Write a program to copy the contents of one text file into another text file.
- 47:** Write a program to demonstrate the use of a simple macro using #define.